CLASS 9 · PART TEST-1 · STUDENT PERFORMANCE REPORT

Pulipati Vikas

Pragya G9 (EXT) Achievers · Adm No **246546650**

Campus HYD/Nallakunta DD Colony
JEE Mains 08 Oct 2026
JEE Advanced 09 Oct 2026

JEE Mains

08 Oct 2026 · 74 questions

170 / 296 — **57.4%**

107	75	8/79	14
ALL-INDIA	STATE	CAMPUS	PROGRAM

Weekly-test average **160.8 / 300** (53.6%), 5 of 6 tests → this test **57.4% ▲ 3.8**

126 marks lost: 110 on 22 wrong answers (incl. 22 negative) · 16 on 4 skipped

JEE Advanced

09 Oct 2026 · 50 questions

60 / 176 — **34.1%**

362	248	28/78	52†
ALL-INDIA	STATE	CAMPUS	PROGRAM

Weekly-test average **79.1 / 180** (44.0%), 5 of 6 tests → this test **34.1% ▼ 9.9**

116 marks lost: 46 on 13 wrong answers (incl. 3 negative) · 67 on 18 skipped · 3 on partly-correct answers

1 Subject scorecard

Your marks in each subject, against the 79 students of this program on campus and against your own weekly-test average.

SUBJECT	MARKS	SCORE VS CAMPUS	SUBJECT AIR	CAMPUS AVG / BEST	WEEKLY AVG	CHANGE	RIGHT / WRONG / SKIP
JEE Mains 170 / 296 · 57.4%							
Maths	**61**/100	61%	**199**	43.1 / 95	**70.6**/100 70.6%	▼ **9.6**	17 / 7 / 1
Physics	**52**/100	52%	**120**	21.9 / 90	**55.2**/100 55.2%	▼ **3.2**	15 / 8 / 2
Chemistry	**57**/96	59%	**156**	41.4 / 81	**34.5**/100 34.5%	▲ **24.8**	16 / 7 / 1
JEE Advanced 60 / 176 · 34.1%							
Maths	**29**/60	48%	**326**	24.5 / 60	**33.0**/60 55.0%	▼ **6.7**	9 / 5 / 3
Physics	**12**/60	20%	**578**	12.3 / 56	**29.0**/60 48.4%	▼ **28.4**	4 / 3 / 10
Chemistry	**19**/56	34%	**441**	15.8 / 37	**16.8**/60 28.0%	▲ **5.9**	6 / 5 / 5

Bar = your score % | campus average | best on campus Change = this test % − weekly-test average %

What this report says

1. **Maths** is your strongest JEE Mains subject: 61/100 (61%), All-India subject rank 199.
2. **Physics** needs the most work in JEE Mains: 52/100 (52%), subject rank 120; you lost 10 of its 25 questions.
3. **Maths** is your strongest JEE Advanced subject: 29/60 (48%), All-India subject rank 326.
4. **Physics** needs the most work in JEE Advanced: 12/60 (20%), subject rank 578; you lost 13 of its 17 questions.
5. Biggest fall against your weekly-test average: **JEE Advanced Physics** – 20% here against 48% on average.
6. Biggest improvement: **JEE Mains Chemistry** – 59% here against 35% on average.
7. **4 questions** you lost were answered correctly by 70%+ of the batch's top 100 – about **18 marks** within easy reach (Mains Q9, Q61, Q65 · Advanced Q40).
8. Negative marking cost you **22 in JEE Mains and 3 in JEE Advanced**.
9. You left **4 in JEE Mains and 18 in JEE Advanced** questions blank, and answered 22 in JEE Mains and 13 in JEE Advanced wrong – both are marks to recover.

JEE Mains – question by question

170 / 296 · 57.4%
08 Oct 2026

2 Every question at a glance

Right · Wrong · Skipped · Deleted (no marks) · Easy miss · Within reach · Tough one

Maths	1	2	3	4	5	**6**	**7**	8	**9**	10	11	12	**13**	14	**15**	16	17	**18**	19	**20**	21	22	23	**24**	25
Physics	26	**27**	28	29	30	**31**	**32**	**33**	34	**35**	36	**37**	**38**	39	40	41	42	**43**	44	**45**	46	47	48	**49**	50
Chemistry	51	52	53	54	55	56	**57**	~~58~~	59	60	**61**	**62**	63	64	**65**	**66**	**67**	68	69	70	**71**	72	73	**74**	75

3 Questions you lost – what to fix first

Every question you got wrong or skipped, one table per subject, easiest marks first. **Top 100** = the batch's top 100 across all campuses; **Campus** = all 79 students of this program on campus.

Easy miss 70%+ of the top 100 got it right – easiest marks to win back · **Within reach** 40–69% · **Tough one** below 40%

Maths

8 of 25 questions lost · **39** marks · ● 1 easy miss · ● 4 within reach · ● 3 tough one

Q	PRIORITY	CHAPTER – WHAT IT TESTS	TYPE	YOU	MARKS LOST	TOP 100 RIGHT	CAMPUS RIGHT
Q9	Easy miss	**Inverse Trigonometric Functions**	Single correct · +4 / −1	**Wrong**	**5**	**87%**	80%
Q13	Within reach	**Trigonometric Equations** – Solutions of tan x + sec x = 2 cos x in [0, 2π]	Single correct · +4 / −1	**Wrong**	**5**	**66%**	58%
Q15	Within reach	**Functions** – Inverse of f(x) = x² + x + 1 on a restricted domain	Single correct · +4 / −1	**Wrong**	**5**	**63%**	44%
Q6	Within reach	**Inverse Trigonometric Functions**	Single correct · +4 / −1	**Wrong**	**5**	**58%**	35%
Q18	Within reach	**Sequences & Series**	Single correct · +4 / −1	**Skipped**	**4**	**50%**	49%
Q7	Tough one	**Trigonometric Equations** – √3 sin πx + cos πx = quadratic – number of solutions	Single correct · +4 / −1	**Wrong**	**5**	**37%**	16%
Q24	Tough one	**Functions** – Even function with GIF, fractional part and signum	Numerical · +4 / −1	**Wrong**	**5**	**20%**	16%
Q20	Tough one	**Inverse Trigonometric Functions** – Composite inverse-trig function – counting solutions	Single correct · +4 / −1	**Wrong**	**5**	**19%**	13%

Physics

10 of 25 questions lost · **48** marks · ● 2 within reach · ● 8 tough one

Q	PRIORITY	CHAPTER – WHAT IT TESTS	TYPE	YOU	MARKS LOST	TOP 100 RIGHT	CAMPUS RIGHT
Q37	Within reach	**Laws of Motion** – Tension in a pulley–block system	Single correct · +4 / −1	**Skipped**	4	**49%**	43%
Q33	Within reach	**Laws of Motion** – Tangential acceleration on a frictionless curved track	Single correct · +4 / −1	**Skipped**	4	**43%**	25%
Q27	Tough one	**Circular Motion** – Coins on a rotating disc – friction limit at each radius	Single correct · +4 / −1	**Wrong**	5	**37%**	23%
Q49	Tough one	**Laws of Motion**	Numerical · +4 / −1	**Wrong**	5	**35%**	34%
Q35	Tough one	**Circular Motion** – Power delivered by the centripetal force	Single correct · +4 / −1	**Wrong**	5	**34%**	18%
Q32	Tough one	**Circular Motion** – Blocks on a rotating frame – ω when sliding starts	Single correct · +4 / −1	**Wrong**	5	**33%**	28%
Q31	Tough one	**Work, Energy & Power** – KE vs height for a stone thrown up from a cliff	Single correct · +4 / −1	**Wrong**	5	**32%**	10%
Q38	Tough one	**Work, Energy & Power** – Work along different paths; conservative or not	Single correct · +4 / −1	**Wrong**	5	**25%**	13%
Q43	Tough one	**Laws of Motion** – Tension in a spring pulled by 5 N at each end	Single correct · +4 / −1	**Wrong**	5	**24%**	10%
Q45	Tough one	**Work, Energy & Power** – Speed on hitting the ground – energy conservation	Single correct · +4 / −1	**Wrong**	5	**18%**	6%

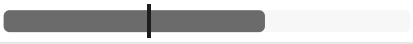
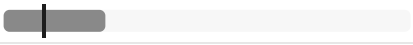
Chemistry

8 of 24 questions lost · **39** marks · ● 2 easy miss · ● 3 within reach · ● 3 tough one

Q	PRIORITY	CHAPTER – WHAT IT TESTS	TYPE	YOU	MARKS LOST	TOP 100 RIGHT	CAMPUS RIGHT
Q61	Easy miss	**Hydrocarbons**	Single correct · +4 / −1	**Skipped**	4	**83%**	78%
Q65	Easy miss	**General Organic Chemistry**	Single correct · +4 / −1	**Wrong**	5	**70%**	62%
Q62	Within reach	**General Organic Chemistry**	Single correct · +4 / −1	**Wrong**	5	**66%**	47%
Q57	Within reach	**Isomerism**	Single correct · +4 / −1	**Wrong**	5	**59%**	42%
Q66	Within reach	**General Organic Chemistry** – False order of −I effect	Single correct · +4 / −1	**Wrong**	5	**45%**	39%
Q67	Tough one	**General Organic Chemistry** – Structure and hybridisation of the methyl carbocation	Single correct · +4 / −1	**Wrong**	5	**26%**	28%
Q74	Tough one	**General Organic Chemistry** – Compounds giving a positive bicarbonate test	Numerical · +4 / −1	**Wrong**	5	**20%**	11%
Q71	Tough one	**Isomerism** – Count of optically active compounds	Numerical · +4 / −1	**Wrong**	5	**6%**	10%

4 Chapter scorecard

Your marks on each chapter's questions against the campus average for the same questions. Weakest chapters first within each subject.

CHAPTER	QS	YOUR MARKS	YOU VS CAMPUS AVERAGE	CAMPUS AVG	QUESTIONS YOU LOST
Maths					
Sequences & Series	1	**0**/4		45%	**Q18**
Inverse Trigonometric Functions	10	**25**/40		48%	**Q6**, **Q9**, **Q20**
Functions	7	**18**/28		35%	**Q15**, **Q24**
Trigonometric Equations	7	**18**/28		44%	**Q7**, **Q13**
Physics					
Work, Energy & Power	5	**5**/20		9%	**Q31**, **Q38**, **Q45**
Laws of Motion	8	**14**/32		23%	**Q33**, **Q37**, **Q43**, **Q49**
Circular Motion	7	**13**/28		30%	**Q27**, **Q32**, **Q35**
Friction	2	**8**/8		17%	– none
Momentum & Collisions	3	**12**/12		25%	– none
Chemistry					
General Organic Chemistry	10	**15**/40		32%	**Q62**, **Q65**, **Q66**, **Q67**, **Q74**
Hydrocarbons	4	**12**/16		42%	**Q61**
Isomerism	10	**30**/40		55%	**Q57**, **Q71**

Bar = your % of the chapter's marks | campus average

5 JEE Mains · Maths – questions to revisit

8 questions · easy misses first
cropped from the question paper

Q9 — Inverse Trigonometric Functions

JEE Mains · Maths · Single correct · marking +4 / −1 · marks lost 5
Top 100: **87%** got it right · Campus: **80%** (63 of 79)

Easy miss
You: Wrong

9. The maximum value of the function $f(x)=\sin^{-1}(x)+\cos^{-1}(x)+\tan^{-1}(x)$

A) equals π B) equals $\frac{3\pi}{4}$ C) equals $\frac{\pi}{4}$ D) does not exist

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q13 — Trigonometric Equations – Solutions of tan x + sec x = 2 cos x in [0, 2π]

JEE Mains · Maths · Single correct · marking +4 / −1 · marks lost 5
Top 100: **66%** got it right · Campus: **58%** (46 of 79)

Within reach
You: Wrong

13. Number of solution(s) of $\tan x+\sec x=2\cos x$ in $[0,2\pi]$ is

A) 1 B) 2 C) 3 D) 4

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q15 — Functions – Inverse of f(x) = x² + x + 1 on a restricted domain

JEE Mains · Maths · Single correct · marking +4 / −1 · marks lost 5
Top 100: **63%** got it right · Campus: **44%** (35 of 79)

Within reach
You: Wrong

15. Let $f:\left(-\infty,\frac{-1}{2}\right]\to\left[\frac{3}{4},\infty\right)$, $f(x)=x^2+x+1$. Then $f^{-1}(x)=\ldots\ldots$

A) $\frac{-1\pm\sqrt{4y-3}}{2}$ B) $\frac{-1\pm\sqrt{4x-3}}{2}$

C) $\frac{-1+\sqrt{4x-3}}{2}$ D) $\frac{-1-\sqrt{4x-3}}{2}$

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q6 — Inverse Trigonometric Functions

JEE Mains · Maths · Single correct · marking +4 / −1 · marks lost 5

Top 100: **58%** got it right · Campus: **35%** (28 of 79)

Within reach

You: Wrong

6. $\tan^{-1}(2)+\tan^{-1}(3)=\ldots\ldots\ldots\ldots$

A) $\left(-\frac{\pi}{4}\right)$ B) $\frac{\pi}{4}$ C) $\frac{3\pi}{4}$ D) $-\frac{3\pi}{4}$

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q18 — Sequences & Series

JEE Mains · Maths · Single correct · marking +4 / −1 · marks lost 4

Top 100: **50%** got it right · Campus: **49%** (39 of 79)

Within reach

You: Skipped

18. Let $f(n)=1^3+2^3+3^3+4^3+\ldots\ldots+n^3$ then the sum $S=1^3+3^3+5^3+\ldots\ldots+(2n-1)^3$ can be written as :

A) $f(2n)+8f(n)$ B) $8f(2n)+f(n)$

C) $f(2n)-8f(n)$ D) $8f(n)-f(2n)$

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q7 — Trigonometric Equations – √3 sin πx + cos πx = quadratic – number of solutions

JEE Mains · Maths · Single correct · marking +4 / −1 · marks lost 5

Top 100: **37%** got it right · Campus: **16%** (13 of 79)

Tough one

You: Wrong

7. Number of solutions of $\sqrt{3}\sin(\pi x)+\cos(\pi x)=x^2-\frac{2}{3}x+\frac{19}{9}$ is ……..

A) 0 B) 1 C) 2 D) 3

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q24 — Functions – Even function with GIF, fractional part and signum

JEE Mains · Maths · Numerical · marking +4 / −1 · marks lost 5

Top 100: **20%** got it right · Campus: **16%** (13 of 79)

Tough one

You: Wrong

24. Let $f(x)=\left([a]^2-5[a]+4\right)x^3-\left(6\{a\}^2-5\{a\}+1\right)x-(\tan x)(\operatorname{sgn}(x))$ (where [.] = GIF, $\{.\}=F.P.F, \operatorname{sgn}(x)=signum(x)$). If $f(x)$ is an even function of 'x', then the sum of possible values of 'a' is ……?

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q20 — Inverse Trigonometric Functions – Composite inverse-trig function – counting solutions

JEE Mains · Maths · Single correct · marking +4 / −1 · marks lost 5

Top 100: **19%** got it right · Campus: **13%** (10 of 79)

Tough one

You: Wrong

20. Consider the functions $rage(x)=\sin^{-1}\sqrt{x}+e^{\left(\tan^{-1}\left(\sqrt{x-1}\right)\right)^2}+\left(\cos^{-1}\left(\sqrt{x}\right)\right)^3$ and $bait(x)=x^2+x+1$ within their respective domains. Then the number of solutions of $rage(bait(x))<3$ is/are ……..

A) 0 B)1 C)2 D)3

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

JEE Mains · Physics – questions to revisit

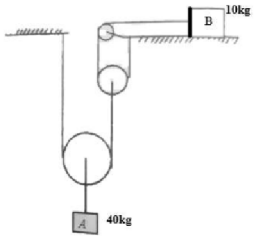
10 questions · easy misses first
cropped from the question paper

Q37 — Laws of Motion – Tension in a pulley-block system

JEE Mains · Physics · Single correct · marking +4 / −1 · marks lost 4
Top 100: **49%** got it right · Campus: **43%** (34 of 79)

Within reach
You: Skipped

37. Fig shows two blocks A and B connected to an ideal pulley string system. In this system when bodies are released then $\left(g = 10m/s^2\right)$(Mass of a is 40Kg and B is 10Kg on smooth surface). The tension in the string connecting the block B is



A) $20N$ B) $80N$ C) $60N$ D) $40N$

Why did I lose this question? (write the reason and the correct approach)

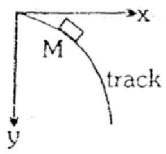
☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q33 — Laws of Motion – Tangential acceleration on a frictionless curved track

JEE Mains · Physics · Single correct · marking +4 / −1 · marks lost 4
Top 100: **43%** got it right · Campus: **25%** (20 of 79)

Within reach
You: Skipped

33. A man M slides down a curved frictionless track, starting from rest. The curve obeys equation $y = \frac{x^2}{2}$. The tangential acceleration of mass is



A) g B) $\frac{gx}{\sqrt{x^2+4}}$ C) $\frac{g}{2}$ D) $\frac{gx}{\sqrt{x^2+1}}$

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q27

Circular Motion – Coins on a rotating disc – friction limit at each radius

JEE Mains · Physics · Single correct · marking +4 / −1 · marks lost 5

Top 100: **37%** got it right · Campus: **23%** (18 of 79)

Tough one

You: Wrong

27. A disc revolves with a speed of 100/3 rev/min and has radius of 15cm. Two coins of same mass as disc are placed at 4 cm and 14cm away from the centre of disc. If the coefficient of friction between the coins and the disc is 0.15, which of the two coins will revolve with disc?

A) both the coins will revolve

B) No coin will revolve with disc

C) coin at 4cm will revolve with disc

D) coin at 14cm will revolve with disc

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q49

Laws of Motion

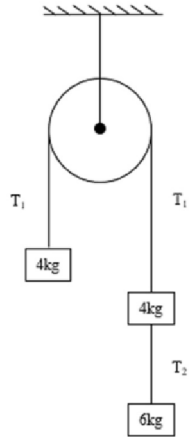
JEE Mains · Physics · Numerical · marking +4 / −1 · marks lost 5

Top 100: **35%** got it right · Campus: **34%** (27 of 79)

Tough one

You: Wrong

49. Assume pulley and string are ideal then find $\frac{T_2}{T_1} = \ldots\ldots\ldots$



Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q35 — Circular Motion – Power delivered by the centripetal force

JEE Mains · Physics · Single correct · marking +4 / −1 · marks lost 5

Top 100: **34%** got it right · Campus: **18%** (14 of 79)

Tough one

You: Wrong

35. A particle of mass m is moving in a circular path of constant radius r such that its centripetal acceleration changing with time 't' as $a_c = k^2t^2$, where K is a constant. The power delivered to the particle by centripetal force acting on it is

A) zero B) mk^2rt C) mk^2r^2t D) $mk^2\sqrt{rt}$

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q32 — Circular Motion – Blocks on a rotating frame – ω when sliding starts

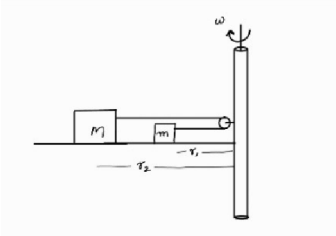
JEE Mains · Physics · Single correct · marking +4 / −1 · marks lost 5

Top 100: **33%** got it right · Campus: **28%** (22 of 79)

Tough one

You: Wrong

32. Two blocks of masses m and M are connected by a chord passing around a frictionless pulley which is attached to a rotating frame, which rotates about a vertical axis with angular velocity ω. If co-efficient of friction between two masses and the surface be μ_1 and μ_2 respectively. The value of ω at which block starts sliding radially $(M > m)$



A) $\omega = \left[g\left[\frac{\mu_1 M + \mu_2 m}{mr_1 - Mr_1} \right] \right]^{1/2}$ B) $\omega = \left[g\left[\frac{\mu_1 m + \mu_2 M}{Mr_2 - mr_1} \right] \right]^{1/2}$

C) $\omega = \left[g\left[\frac{\mu_1 m - \mu_2 M}{Mr_2 - mr_1} \right] \right]^{1/2}$ D) $\omega = \left[g\left[\frac{\mu_1 M - \mu_2 m}{Mr_2 - mr_1} \right] \right]^{1/2}$

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q31 **Work, Energy & Power – KE vs height for a stone thrown up from a cliff**

JEE Mains · Physics · Single correct · marking +4 / −1 · marks lost 5

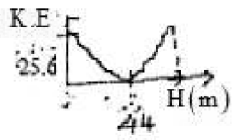
Top 100: **32%** got it right · Campus: **10%** (8 of 79)

Tough one

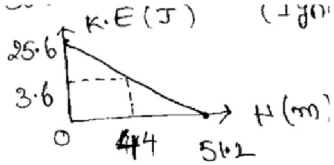
You: Wrong

31. A stone of mass 50g is thrown vertically upwards with a velocity of 12m/s from edge of a cliff 44m high. Identify kinetic energy of stone versus height from bottom of cliff $\left(g = 10m/s^2\right)$ (ignore air resistance)

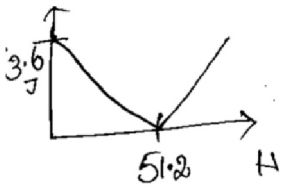
A)



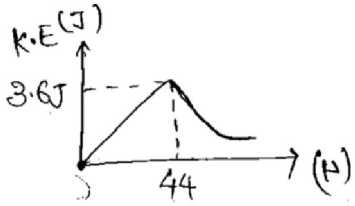
B)



C)



D)



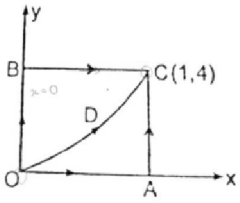
Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q38 **Work, Energy & Power** – **Work along different paths; conservative or not**
JEE Mains · Physics · Single correct · marking +4 / −1 · marks lost 5
Top 100: **25%** got it right · Campus: **13%** (10 of 79)

Tough one
You: Wrong

38. A particle is moved along the different paths OAC, OBC and ODC as shown in the fig. path ODC is parabola, $y = 4x^2$. The work done by a force $\vec{F} = xy\hat{i} + x^2 y\hat{j}$ on the particle along



Column – I	**Column – II**
a)Path OAC	P) 2J
b) Path OBC	Q) $\frac{19}{3}$J
c) Path ODC	R) Conservative
d) Force is	S) 8J
	T) Non- conservative

A) $a \to Q, b \to S, c \to P, d \to T$ B) $a \to S, b \to P, c \to Q, d \to T$

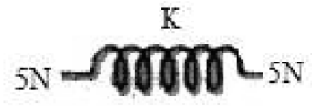
C) $a \to S, b \to Q, c \to P, d \to T$ D) $a \to S, b \to Q, c \to S, d \to T$

Why did I lose this question? (write the reason and the correct approach)
☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q43 **Laws of Motion** – **Tension in a spring pulled by 5 N at each end**
JEE Mains · Physics · Single correct · marking +4 / −1 · marks lost 5
Top 100: **24%** got it right · Campus: **10%** (8 of 79)

Tough one
You: Wrong

43. The tension in the massless spring is



A) zero B) 10N C) 2.5N D) 5N

Why did I lose this question? (write the reason and the correct approach)
☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q45 **Work, Energy & Power – Speed on hitting the ground – energy conservation**
JEE Mains · Physics · Single correct · marking +4 / −1 · marks lost 5
Top 100: **18%** got it right · Campus: **6%** (5 of 79)

Tough one
You: Wrong

45. A cricket bowler releases the ball in two different ways

i) giving it only horizontal velocity and

ii) giving it horizontal velocity and small downward velocity

The speeds at the time release is same. Both are released at a height H from ground then which one will have greater speed when ball hits the ground?(Neglect air resistance)

A) Ist way B) IInd way C) same in two ways D) None of the above

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

JEE Mains · Chemistry – questions to revisit

8 questions · easy misses first
cropped from the question paper

Q61 Hydrocarbons

JEE Mains · Chemistry · Single correct · marking +4 / −1 · marks lost 4
Top 100: **83%** got it right · Campus: **78%** (62 of 79)

Easy miss
You: Skipped

61. Which of the following statement is /are true of catalytic hydrogenation reaction

A) Dihydrogen gas adds to alkenes and alkynes in the presence of finely divided catalysts like Pt, Pd or Ni to form alkanes

B) The metal adsorbs dihydrogen gas on their surfaces and breaks the hydrogen – hydrogen bond

C) Pt and Pd catalyse the reaction at room temperature but relatively higher temperature and pressure are required with nickel catalyst

D) All of these

Why did I lose this question? (write the reason and the correct approach)

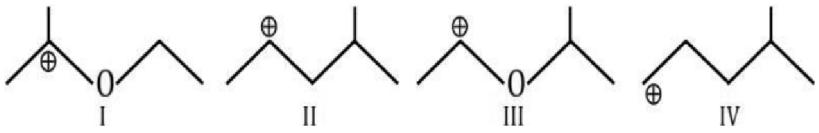
☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q65 General Organic Chemistry

JEE Mains · Chemistry · Single correct · marking +4 / −1 · marks lost 5
Top 100: **70%** got it right · Campus: **62%** (49 of 79)

Easy miss
You: Wrong

65. The correct stability order of the following species is :



A) II > IV > I > III

B) I > II > III > IV

C) II > I> IV > III

D) I > III > II > IV

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q62

General Organic Chemistry

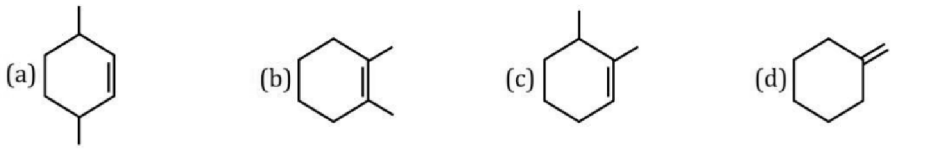
JEE Mains · Chemistry · Single correct · marking +4 / −1 · marks lost 5

Top 100: **66%** got it right · Campus: **47%** (37 of 79)

Within reach

You: Wrong

62. Decreasing order of heat evolved upon catalytic hydrogenation of given reaction with a H_2 / Pd is (only based on hyper conjugation effect)



(A) b > c > a > d (B) a > d > c > b (C) d > c > a > b (D) c > b > c > d

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q57

Isomerism

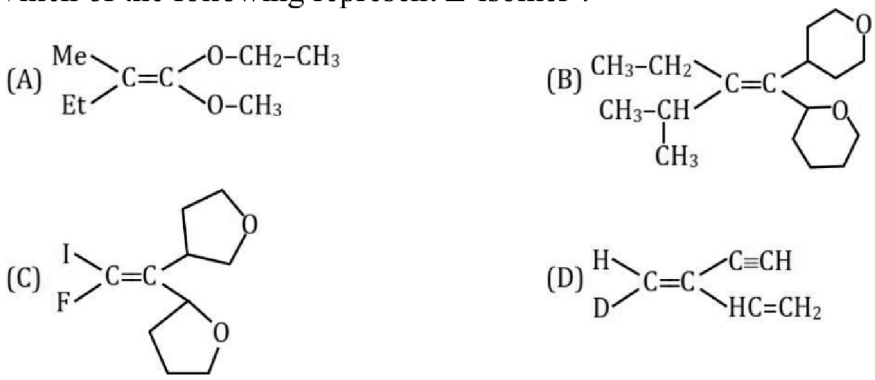
JEE Mains · Chemistry · Single correct · marking +4 / −1 · marks lost 5

Top 100: **59%** got it right · Campus: **42%** (33 of 79)

Within reach

You: Wrong

57. Which of the following represent Z-isomer ?



Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q66

General Organic Chemistry – **False order of –I effect**

JEE Mains · Chemistry · Single correct · marking +4 / −1 · marks lost 5

Top 100: **45%** got it right · Campus: **39%** (31 of 79)

Within reach

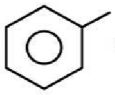
You: Wrong

66. Which of the following is false order of – I effect ?

A) $-F > -Cl > -Br > -I$

B) $-\overset{\oplus}{N}R_3 > -\overset{\oplus}{N}H_3 > -NO_2$

C) $-F > -OH > -NH_2$

D)  $> -C \equiv CH > -H$

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q67

General Organic Chemistry – **Structure and hybridisation of the methyl carbocation**

JEE Mains · Chemistry · Single correct · marking +4 / −1 · marks lost 5

Top 100: **26%** got it right · Campus: **28%** (22 of 79)

Tough one

You: Wrong

67. Wrong statement regarding methyl carbonium ion $\overset{\oplus}{C}H_3$

A) It is sp^2 hybridized

B) Vacant orbital is sp^2 hybridized

C) vacant orbital is perpendicular to molecular planar and is pure p-orbital

D) It is electrophilic with sextet of electron

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q74

General Organic Chemistry – **Compounds giving a positive bicarbonate test**

JEE Mains · Chemistry · Numerical · marking +4 / −1 · marks lost 5

Top 100: **20%** got it right · Campus: **11%** (9 of 79)

Tough one

You: Wrong

74. How many of the following compounds can give positive bicarbonate test?

HCl, Picric Acid, Benzoic acid, 2,4-dintrophenol, 4-nitrophenol, Squaric Acid, formic Acid, Benzene Sulphuric acid, 1-Butyne, NH_4Cl .

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q71 **Isomerism – Count of optically active compounds**

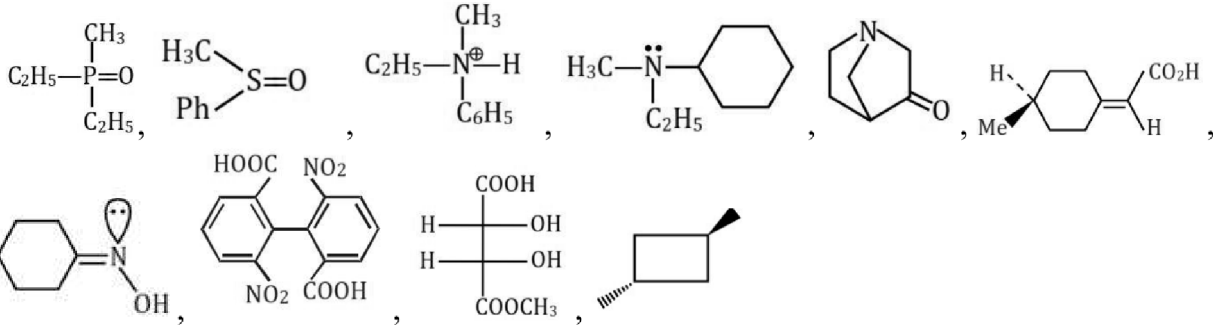
JEE Mains · Chemistry · Numerical · marking +4 / −1 · marks lost 5

Top 100: **6%** got it right · Campus: **10%** (8 of 79)

Tough one

You: Wrong

71. Total no of compounds which are optically active are _____



Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

6 Every question at a glance

Right · Wrong · Skipped · Deleted (no marks) · Easy miss · Within reach · Tough one

Maths	1	2	3	4	5	6	7	**8**	**9**	10	11	**12**	**13**	**14**	**15**	**16**	**17**
Physics	**18**	**19**	20	21	**22**	**23**	24	**25**	**26**	**27**	**28**	**29**	**30**	31	**32**	**33**	**34**
Chemistry	**35**	**36**	37	38	39	**40**	**41**	**42**	43	44	~~45~~	**46**	**47**	48	**49**	**50**	**51**

7 Questions you lost – what to fix first

Every question you got wrong or skipped, one table per subject, easiest marks first. **Top 100** = the batch's top 100 across all campuses; **Campus** = all 78 students of this program on campus.

Easy miss 70%+ of the top 100 got it right – easiest marks to win back · **Within reach** 40–69% · **Tough one** below 40%

Maths

8 of 17 questions lost · **28** marks (+3 on partly-correct answers) · 6 within reach · 2 tough one

Q	PRIORITY	CHAPTER – WHAT IT TESTS	TYPE	YOU	MARKS LOST	TOP 100 RIGHT	CAMPUS RIGHT
Q14	Within reach	**Inverse Trigonometric Functions**	Paragraph · +3 / 0	**Wrong**	**3**	**63%**	58%
Q15	Within reach	**Inverse Trigonometric Functions**	Paragraph · +3 / 0	**Wrong**	**3**	**57%**	53%
Q12	Within reach	**Functions**	Integer · +4 / 0	**Wrong**	**4**	**55%**	40%
Q9	Within reach	**Inverse Trigonometric Functions**	Integer · +4 / 0	**Wrong**	**4**	**54%**	58%
Q13	Within reach	**Inverse Trigonometric Functions**	Integer · +4 / 0	**Wrong**	**4**	**54%**	45%
Q16	Within reach	**Trigonometric Equations**	Paragraph · +3 / 0	**Skipped**	**3**	**51%**	42%
Q8	Tough one	**Functions** – Period from two lines of symmetry	Integer · +4 / 0	**Skipped**	**4**	**12%**	13%
Q17	Tough one	**Trigonometric Equations** – Sum of solutions of $\cos(\pi\sqrt{(x-100)})\cdot\cos(\pi\sqrt{x}) = 1$	Paragraph · +3 / 0	**Skipped**	**3**	**6%**	6%

Physics

13 of 17 questions lost · **48** marks · ● 4 within reach · ● 9 tough one

Q	PRIORITY	CHAPTER – WHAT IT TESTS	TYPE	YOU	MARKS LOST	TOP 100 RIGHT	CAMPUS RIGHT
Q23	Within reach	**Laws of Motion**	Multiple correct · +4 / −2	**Skipped**	4	**56%**	51%
Q28	Within reach	**Momentum & Collisions**	Integer · +4 / 0	**Skipped**	4	**49%**	33%
Q25	Within reach	**Laws of Motion**	Integer · +4 / 0	**Skipped**	4	**47%**	44%
Q19	Within reach	**Circular Motion**	Single correct · +3 / −1	**Skipped**	3	**40%**	38%
Q22	Tough one	**Laws of Motion**	Multiple correct · +4 / −2	**Skipped**	4	**36%**	38%
Q26	Tough one	**Circular Motion**	Integer · +4 / 0	**Skipped**	4	**24%**	17%
Q29	Tough one	**Circular Motion** – Net acceleration in a vertical circle at 37°	Integer · +4 / 0	**Skipped**	4	**19%**	14%
Q30	Tough one	**Friction** – Work by friction seen from an accelerating observer	Integer · +4 / 0	**Skipped**	4	**19%**	12%
Q18	Tough one	**Friction** – Static vs kinetic friction under a time-varying force	Single correct · +3 / −1	**Wrong**	4	**16%**	9%
Q34	Tough one	**Work, Energy & Power** – Work by conservative forces from the U–x graph	Paragraph · +3 / 0	**Wrong**	3	**12%**	9%
Q27	Tough one	**Friction** – Net work done by friction between sliding blocks	Integer · +4 / 0	**Skipped**	4	**8%**	10%
Q32	Tough one	**Friction** – Total force exerted by a block on an accelerating cube	Paragraph · +3 / 0	**Skipped**	3	**8%**	6%
Q33	Tough one	**Work, Energy & Power** – Work-energy theorem from a–t and U–x graphs	Paragraph · +3 / 0	**Wrong**	3	**2%**	1%

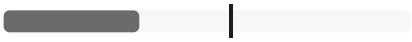
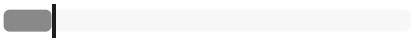
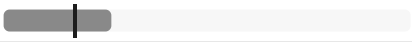
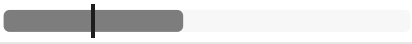
Chemistry

10 of 16 questions lost · **37** marks · ● 1 easy miss · ● 4 within reach · ● 5 tough one

Q	PRIORITY	CHAPTER – WHAT IT TESTS	TYPE	YOU	MARKS LOST	TOP 100 RIGHT	CAMPUS RIGHT
Q40	Easy miss	**General Organic Chemistry**	Multiple correct · +4 / −2	**Skipped**	4	**79%**	73%
Q36	Within reach	**General Organic Chemistry**	Single correct · +3 / −1	**Wrong**	4	**64%**	53%
Q35	Within reach	**General Organic Chemistry**	Single correct · +3 / −1	**Wrong**	4	**58%**	42%
Q49	Within reach	**General Organic Chemistry**	Paragraph · +3 / 0	**Wrong**	3	**53%**	46%
Q41	Within reach	**Isomerism**	Multiple correct · +4 / −2	**Skipped**	4	**43%**	42%
Q42	Tough one	**General Organic Chemistry**	Integer · +4 / 0	**Skipped**	4	**39%**	36%
Q46	Tough one	**Isomerism** – Count of optically active compounds	Integer · +4 / 0	**Skipped**	4	**15%**	15%
Q51	Tough one	**Hydrocarbons** – Chiral products of free-radical monobromination	Paragraph · +3 / 0	**Wrong**	3	**15%**	4%
Q47	Tough one	**Hydrocarbons** – Dichlorination isomers of C_5H_{10} (VD 35)	Integer · +4 / 0	**Wrong**	4	**7%**	5%
Q50	Tough one	**Hydrocarbons** – Preparation of alkanes – count correct statements	Paragraph · +3 / 0	**Skipped**	3	**4%**	3%

8 Chapter scorecard

Your marks on each chapter's questions against the campus average for the same questions. Weakest chapters first within each subject. Chapter marks are approximate (≈): partly-correct answers cannot be traced to a question.

CHAPTER	QS	YOUR MARKS	YOU VS CAMPUS AVERAGE	CAMPUS AVG	QUESTIONS YOU LOST
Maths					
Inverse Trigonometric Functions	6	≈ **7**/21		55%	**Q9**, **Q13**, **Q14**, **Q15**
Functions	6	≈ **14**/22		40%	**Q8**, **Q12**
Trigonometric Equations	5	≈ **11**/17		35%	**Q16**, **Q17**
Physics					
Laws of Motion	3	≈ **0**/12		39%	**Q22**, **Q23**, **Q25**
Momentum & Collisions	1	≈ **0**/4		33%	**Q28**
Friction	5	≈ **2**/17		12%	**Q18**, **Q27**, **Q30**, **Q32**
Circular Motion	4	≈ **4**/15		17%	**Q19**, **Q26**, **Q29**
Work, Energy & Power	4	≈ **6**/12		22%	**Q33**, **Q34**
Chemistry					
Isomerism	2	≈ **0**/8		16%	**Q41**, **Q46**
General Organic Chemistry	9	≈ **11**/30		43%	**Q35**, **Q36**, **Q40**, **Q42**, **Q49**
Hydrocarbons	5	≈ **8**/18		22%	**Q47**, **Q50**, **Q51**

Bar = your % of the chapter's marks | campus average

Q14

Inverse Trigonometric Functions

JEE Advanced · Maths · Paragraph · marking +3 / 0 · marks lost 3

Top 100: **63%** got it right · Campus: **58%** (45 of 78)

Within reach

You: Wrong

PARAGRAPH-I

A mischievous student decides to play around with inverse trigonometric functions. She decides that since $f:\left[\frac{\pi}{2},\frac{3\pi}{2}\right]\to[-1,1], f(x)=\sin x$ and $g:[\pi,2\pi]\to[-1,1], g(x)=\cos x$ are invertible , she defines

$$h:[-1,1]\to\left[\frac{\pi}{2},\frac{3\pi}{2}\right], h(x)=\sin^{-1}x$$

$$j:[-1,1]\to[\pi,2\pi], j(x)=\cos^{-1}x$$

So that $h=f^{-1}$ and $j=g^{-1}$. (For example under this definition $\sin^{-1}(0)=\pi$ and $\cos^{-1}(0)=\frac{3\pi}{2}$). Under these definitions, answer the following :(use $\pi=3.14$)

14. $\sin^{-1}(1)+\cos^{-1}(1)=.....$

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q15

Inverse Trigonometric Functions
JEE Advanced · Maths · Paragraph · marking +3 / 0 · marks lost 3
Top 100: **57%** got it right · Campus: **53%** (41 of 78)

Within reach
You: Wrong

PARAGRAPH-I

A mischievous student decides to play around with inverse trigonometric functions. She decides that since $f:\left[\frac{\pi}{2},\frac{3\pi}{2}\right]\to[-1,1], f(x)=\sin x$ and $g:[\pi,2\pi]\to[-1,1], g(x)=\cos x$ are invertible , she defines

$$h:[-1,1]\to\left[\frac{\pi}{2},\frac{3\pi}{2}\right], h(x)=\sin^{-1}x$$

$$j:[-1,1]\to[\pi,2\pi], j(x)=\cos^{-1}x$$

So that $h=f^{-1}$ and $j=g^{-1}$. (For example under this definition $\sin^{-1}(0)=\pi$ and $\cos^{-1}(0)=\frac{3\pi}{2}$). Under these definitions, answer the following :(use $\pi=3.14$)

15. $\sin^{-1}\left(\sin\frac{\pi}{3}\right)=\ldots\ldots\ldots\ldots$

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q12

Functions
JEE Advanced · Maths · Integer · marking +4 / 0 · marks lost 4
Top 100: **55%** got it right · Campus: **40%** (31 of 78)

Within reach
You: Wrong

12. Consider the function $f(x)=\log_{([x-1])}\left(\frac{|x|}{x}\right), ([.]=GIF)$

Let 'n' be the smallest positive integer in the domain of f . Let 'S' be the sum of all integers in the range of f . Then $n+S=\ldots\ldots\ldots$

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q9 Inverse Trigonometric Functions

JEE Advanced · Maths · Integer · marking +4 / 0 · marks lost 4

Top 100: **54%** got it right · Campus: **58%** (45 of 78)

Within reach

You: Wrong

9. $\sum_{n=1}^{5} \sin^{-1}\left(\sin\left(2n-1\right)\right) = \ldots\ldots\ldots\ldots\ldots$

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q13 Inverse Trigonometric Functions

JEE Advanced · Maths · Integer · marking +4 / 0 · marks lost 4

Top 100: **54%** got it right · Campus: **45%** (35 of 78)

Within reach

You: Wrong

13. If $\cos^{-1}\left(x_1\right)+\cos^{-1}\left(x_2\right)+\cos^{-1}\left(x_3\right)+\ldots\ldots\ldots+\cos^{-1}\left(x_{2026}\right)=0$, then $x_2+x_4+x_6+\ldots\ldots+x_{2026}=\ldots\ldots\ldots\ldots$

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q16 Trigonometric Equations

JEE Advanced · Maths · Paragraph · marking +3 / 0 · marks lost 3

Top 100: **51%** got it right · Campus: **42%** (33 of 78)

Within reach

You: Skipped

PARAGRAPH-II

Given positive integer 'k' , if we want to find positive integer 'n' and 'm' such that $n^2-m^2=k$ then we use $\left(n+m\right)\left(n-m\right)=k$ where we equate $n+m$ & $n-m$ to factors of k to find all possible values of $\left(n,m\right)$.

Eg: $n^2-m^2=6\Rightarrow\left(n+m\right)\left(n-m\right)=\left(\pm6\right)\left(\pm1\right)$ or $\left(\pm3\right)\left(\pm2\right)$

16. The sum of real solution(s) of $\cos\left(\pi\sqrt{x-4}\right)\cos\left(\pi\sqrt{x}\right)=1$ is $\ldots\ldots\ldots\ldots$

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q8

Functions – Period from two lines of symmetry

JEE Advanced · Maths · Integer · marking +4 / 0 · marks lost 4

Top 100: **12%** got it right · Campus: **13%** (10 of 78)

Tough one

You: Skipped

8. Let $a \in R$. If $f(a-x)=f(a+x)\forall x \in R$, then $x=a$ is a line of symmetry in the graph of $y=f(x)$. Consider a non-constant function $g(x)$ such that $g(1+x)=g(1-x)\forall x \in R$ and $g(2026+x)=g(2026-x)\forall x \in R$. Then all such $g(x)$ must be periodic. The smallest positive number that could be a period for all such $g(x)$ is .....

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q17

Trigonometric Equations – Sum of solutions of cos(π√(x−100))·cos(π√x) = 1

JEE Advanced · Maths · Paragraph · marking +3 / 0 · marks lost 3

Top 100: **6%** got it right · Campus: **6%** (5 of 78)

Tough one

You: Skipped

PARAGRAPH-II

Given positive integer 'k' , if we want to find positive integer 'n' and 'm' such that $n^2-m^2=k$ then we use $(n+m)(n-m)=k$ where we equate $n+m$ & $n-m$ to factors of k to find all possible values of (n,m).

Eg: $n^2-m^2=6 \Rightarrow (n+m)(n-m)=(\pm 6)(\pm 1)$ or $(\pm 3)(\pm 2)$

17. The sum of real solution(s) of $\cos\left(\pi\sqrt{x-100}\right)\cos\left(\pi\sqrt{x}\right)=1$ is .....

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

JEE Advanced · Physics – questions to revisit

13 questions · easy misses first
cropped from the question paper

Q23 Laws of Motion

JEE Advanced · Physics · Multiple correct · marking +4 / −2 · marks lost 4
Top 100: **56%** got it right · Campus: **51%** (40 of 78)

Within reach
You: Skipped

23. An object of mass m is at rest in equilibrium at the origin. At $t = 0\text{sec}$ a force $\vec{F}(t)$ is applied which has the components $F_x(y) = k_1 + k_2 y$ $F_y(t) = k_3 t$ where k_1, k_2, k_3 are constants . Then

A) X-component of velocity at any time 't' is $V_x = \frac{k_1 t}{m} + \frac{k_2 k_3}{24m^2} t^4$

B) X-component of acceleration at anytime 't' is $a_x = \left(\frac{k_1}{m} + \frac{k_2 k_3}{6m^2} t^3 \right)$

C) Y- coordinate of object at any time 't' is $\frac{k_3 t^2}{3m}$

D) Y- coordinate of object at any time 't' is $\frac{k_3 t^3}{6m}$

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q28 Momentum & Collisions

JEE Advanced · Physics · Integer · marking +4 / 0 · marks lost 4
Top 100: **49%** got it right · Campus: **33%** (26 of 78)

Within reach
You: Skipped

28. Three objects A, B and C are kept in a straight line on a frictionless horizontal surfaces. These have masses m,2m and m respectively. The object A moves towards B with speed 9m/s and makes an elastic collision with it. There after B makes completely inelastic collision with 'C'. All motion occur on same straight line final velocity of object 'C' is ……m/s.

Why did I lose this question? (write the reason and the correct approach)

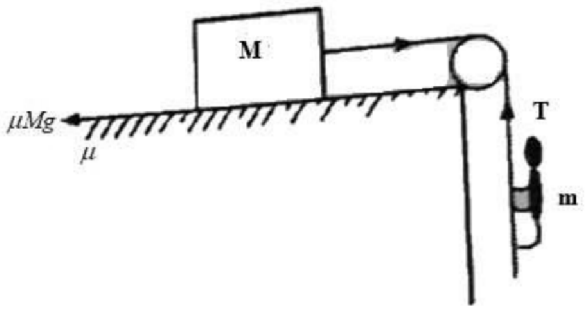
☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q25

Laws of Motion
JEE Advanced · Physics · Integer · marking +4 / 0 · marks lost 4
Top 100: **47%** got it right · Campus: **44%** (34 of 78)

Within reach
You: Skipped

25. A monkey of mass 'm' climbs up to a rope hung over a fixed pulley with an acceleration relative to rope is $\frac{g}{4} m/s^2$. The opposite end of the rope is tied to a block of mass M lying on a rough horizontal plane. The coefficient of friction between the block and horizontal plane is μ. Tension in rope is $T = \frac{M[5m - 4\mu M]}{k[M+m]} + \mu Mg$ value of k is .....



Why did I lose this question? (write the reason and the correct approach)

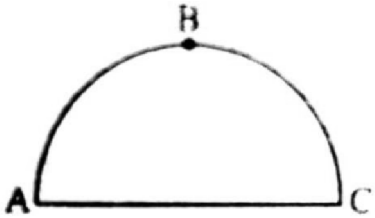
☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q19

Circular Motion
JEE Advanced · Physics · Single correct · marking +3 / −1 · marks lost 3
Top 100: **40%** got it right · Campus: **38%** (30 of 78)

Within reach
You: Skipped

19. If a particle starts from 'A' moving along the curved semi circular path as shown in figure with tangential acceleration 'a' then acceleration at B in magnitude is



A) $2a\left[\sqrt{1+\pi^2}\right]$ B) $a\left[\sqrt{1+\pi^2}\right]$ C) $a\left[\sqrt{\pi^2-1}\right]$ D) $a\pi\left[\sqrt{1+\pi^2}\right]$

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q22

Laws of Motion

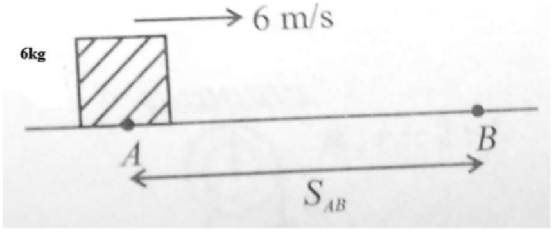
JEE Advanced · Physics · Multiple correct · marking +4 / −2 · marks lost 4

Top 100: **36%** got it right · Campus: **38%** (30 of 78)

Tough one

You: Skipped

22. A block of mass 6kg is projected along the horizontal surface. A resistance force $F=-2v^3$ is acting on the block through out the motion. During motion when it crosses point A and B it speeds are 6m/s and 2m/s respectively. Then



A) distance from A to B $S_{AB}=\frac{2}{3}m$ B) distance from A to B $S_{AB}=1m$

C) Time taken from A to B $T_{AB}=\frac{1}{3}\sec$ D) time taken from A to B $T_{AB}=\frac{2}{3}\sec$

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q26

Circular Motion

JEE Advanced · Physics · Integer · marking +4 / 0 · marks lost 4

Top 100: **24%** got it right · Campus: **17%** (13 of 78)

Tough one

You: Skipped

26. A particle of mass 2.5kg is moving in gravity free space with velocity $3\hat{i}m/s$. At $t=0\sec$ a force of magnitude 67.5N starts acting on the particle such that it is always perpendicular to instantaneous velocity. The minimum time after which particle has same velocity as the initial is $\frac{2\pi}{x}\sec$. The value of x .....

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q29

Circular Motion – Net acceleration in a vertical circle at 37°

JEE Advanced · Physics · Integer · marking +4 / 0 · marks lost 4

Top 100: **19%** got it right · Campus: **14%** (11 of 78)

Tough one

You: Skipped

29. A small ball of mass m is attached to the end of string of length $l = 1m$ whose other end is fixed. From its lowest position, the ball is given a kinetic energy $\frac{mgl}{5}$. Net acceleration $\left(in\ m/s^2\right)$ of ball at the instant when string makes an angle $\theta=37^o$ with vertical is ..... $\left(g = 10m/s^2\right)$

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q30

Friction – Work by friction seen from an accelerating observer

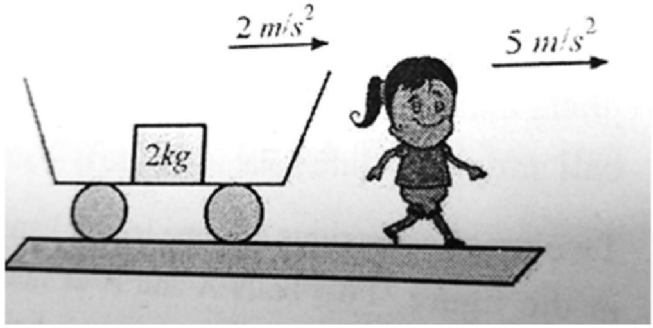
JEE Advanced · Physics · Integer · marking +4 / 0 · marks lost 4

Top 100: **19%** got it right · Campus: **12%** (9 of 78)

Tough one

You: Skipped

30. An observer and a vehicle, both starts moving together from rest with acceleration $5m/s^2$ and $2m/s^2$ respectively. There is a 2 kg block on the floor of the vehicle and $\mu = 0.3$ between their surfaces. The work done by the frictional force on 2kg block as observed by running observer , during first 2seconds of motion is –x (J) $\left(g = 10m/s^2\right)$ find 'x'



Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q18 **Friction** – Static vs kinetic friction under a time-varying force

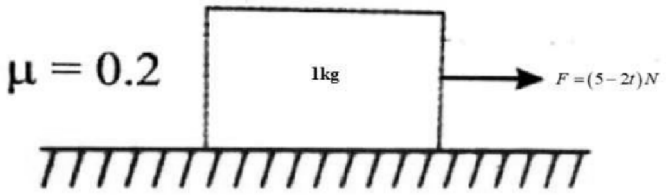
JEE Advanced · Physics · Single correct · marking +3 / −1 · marks lost 4

Top 100: **16%** got it right · Campus: **9%** (7 of 78)

Tough one

You: Wrong

18. The force acting on the block of mass 1 kg is given by $F = 5 - 2t$. The frictional force acting on the block after time $t = 2\text{sec}$ will be $\left(\mu = 0.2, g = 10m/s^2\right)$



A) 2N B) 3N C) 1N D) zero

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q34 **Work, Energy & Power** – Work by conservative forces from the U-x graph

JEE Advanced · Physics · Paragraph · marking +3 / 0 · marks lost 3

Top 100: **12%** got it right · Campus: **9%** (7 of 78)

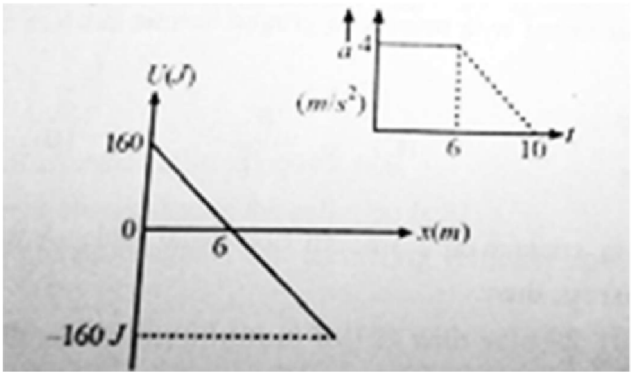
Tough one

You: Wrong

PARAGRAPH-II

A particle of mass 1kg is moving along x-axis its velocity is 6m/s at x=0. Acceleration versus time and P.E versus displacement of particle are shown $\left(g = 10m/s^2\right)$

Acceleration a is in $m/\sec^2$ b



34. Work done by conservative forces _____ J

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q27

Friction – Net work done by friction between sliding blocks

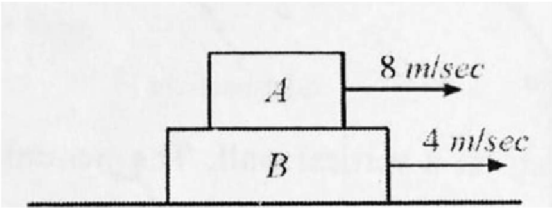
JEE Advanced · Physics · Integer · marking +4 / 0 · marks lost 4

Top 100: **8%** got it right · Campus: **10%** (8 of 78)

Tough one

You: Skipped

27. Block A of mass 1kg is placed on rough surface of block B of mass 3kg. Block B is placed on smooth horizontal surface. Blocks are given velocities as shown. If net workdone by frictional force is "-n" J. value of n is ……..



Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q32

Friction – Total force exerted by a block on an accelerating cube

JEE Advanced · Physics · Paragraph · marking +3 / 0 · marks lost 3

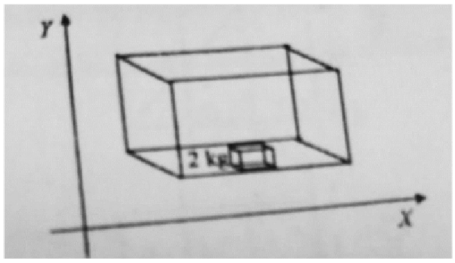
Top 100: **8%** got it right · Campus: **6%** (5 of 78)

Tough one

You: Skipped

PARAGRAPH-I

A solid block of mass 2 kg is resting inside a cube as shown in figure. The cube is moving with a velocity $\vec{V}=\left(5t\hat{i}+2t\hat{j}\right)m/s$. Here 't' is second. The block is in rest w.r.t cube and $\mu=0.5$ between block and surface of cube. $\left(g=10m/s^2\right)$



32. Total force exerted by block on cube is _____ N

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

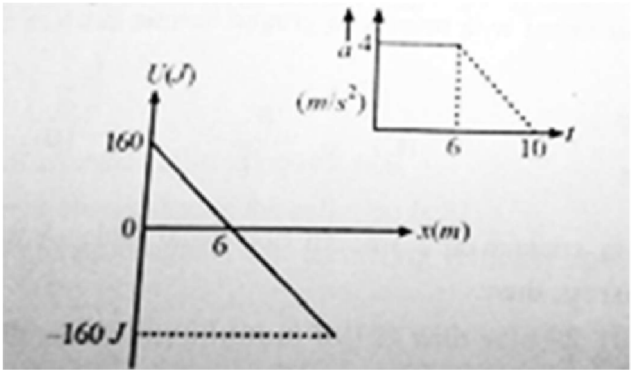
Q33 **Work, Energy & Power** – **Work-energy theorem from a-t and U-x graphs**
JEE Advanced · Physics · Paragraph · marking +3 / 0 · marks lost 3
Top 100: **2%** got it right · Campus: **1%** (1 of 78)

Tough one
You: Wrong

PARAGRAPH-II

A particle of mass 1kg is moving along x-axis its velocity is 6m/s at x=0. Acceleration versus time and P.E versus displacement of particle are shown $\left(g = 10 m/s^2\right)$

Acceleration a is in $m/\sec^2$ b



33. Work done by all forces _____ J

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

JEE Advanced · Chemistry – questions to revisit

10 questions · easy misses first
cropped from the question paper

Q40

General Organic Chemistry
JEE Advanced · Chemistry · Multiple correct · marking +4 / −2 · marks lost 4
Top 100: **79%** got it right · Campus: **73%** (57 of 78)

Easy miss
You: Skipped

40. Which among the following statements are correct?

A) Energy needed for homolytic bond fission is less than that required for the heterolytic bond fission

B) Homolytic bond fission gives neutral species which is paramagnetic in nature

C) Homolytic bond fission takes place in polar solvents

D) Cation and anion is produced by heterolytic bond fission

Why did I lose this question? (write the reason and the correct approach)

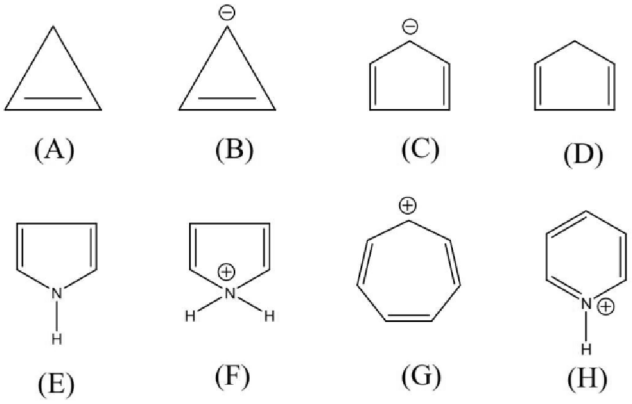
☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q36

General Organic Chemistry
JEE Advanced · Chemistry · Single correct · marking +3 / −1 · marks lost 4
Top 100: **64%** got it right · Campus: **53%** (41 of 78)

Within reach
You: Wrong

36. Designate whether each of the following compound is aromatic (or) not aromatic



A) A,E,G,H are aromatic and B,C,D,F are not aromatic

B) A,C,E,G,H are aromatic and B,D F are not aromatic

C) A,C,D,E,G are aromatic and B,F,H are not aromatic

C) C,E,G,H are aromatic and A,B,D,F are not aromatic

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q35

General Organic Chemistry

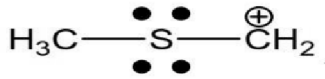
JEE Advanced · Chemistry · Single correct · marking +3 / −1 · marks lost 4

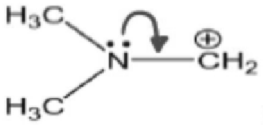
Top 100: **58%** got it right · Campus: **42%** (33 of 78)

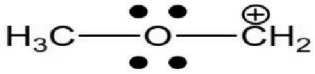
Within reach

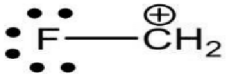
You: Wrong

35. Which of the following cation would have greatest stability?

A) 

B) 

C) 

D) 

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q49

General Organic Chemistry

JEE Advanced · Chemistry · Paragraph · marking +3 / 0 · marks lost 3

Top 100: **53%** got it right · Campus: **46%** (36 of 78)

Within reach

You: Wrong

PARAGRAPH-I

Electrophiles and nucleophiles are fundamental concepts of organic chemistry playing crucial roles in various reactions. Electron loving species that accept electron pairs to form covalent bonds are electrophiles, Nucleus loving species that donate electron pair to form covalent bonds.

49. How many of the following are nucleophiles

A) OR^- B) CH_3^+ C) NH_3 D) $AlCl_3$

E) NO_2^- F) NO_2^+ G) Br^+

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q41

Isomerism

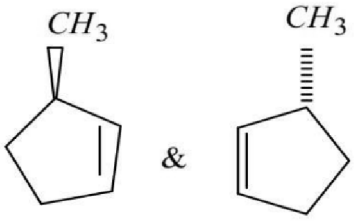
JEE Advanced · Chemistry · Multiple correct · marking +4 / −2 · marks lost 4

Top 100: **43%** got it right · Campus: **42%** (33 of 78)

Within reach

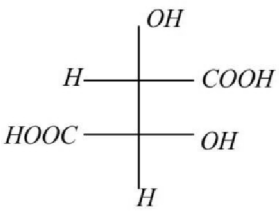
You: Skipped

41. Select the correct statements:



A) are enantiomers

B) The Boiling point or E-isomers is more than Z-isomers of 1-chloro propene



C) is optically active & may be dextro rotatory compound

D) Molecular formula C_3H_6O, possible to show geometrical isomerism.

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q42

General Organic Chemistry

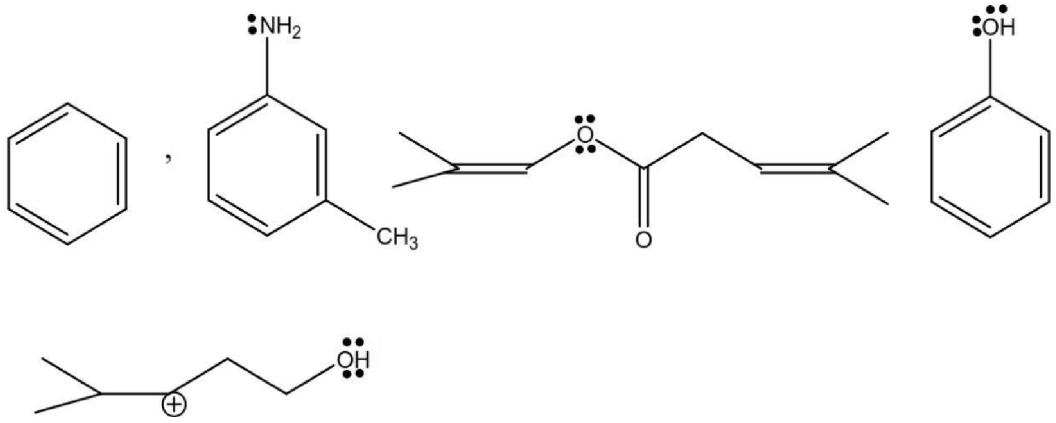
JEE Advanced · Chemistry · Integer · marking +4 / 0 · marks lost 4

Top 100: **39%** got it right · Campus: **36%** (28 of 78)

Tough one

You: Skipped

42. How many compounds among the following show inductive, mesomeric as well as hyper conjugation effects



Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q46

Isomerism – Count of optically active compounds

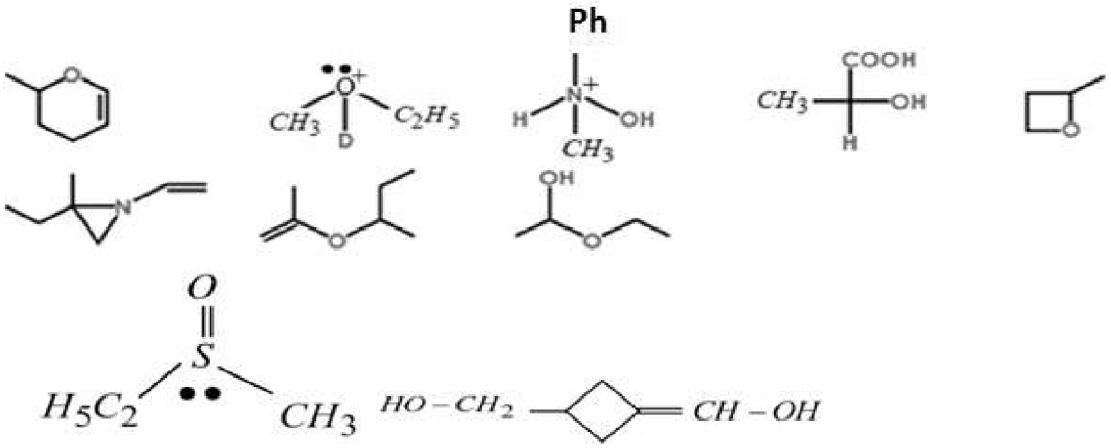
JEE Advanced · Chemistry · Integer · marking +4 / 0 · marks lost 4

Top 100: **15%** got it right · Campus: **15%** (12 of 78)

Tough one

You: Skipped

46. How many given compounds are optically active



Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q51

Hydrocarbons – Chiral products of free-radical monobromination

JEE Advanced · Chemistry · Paragraph · marking +3 / 0 · marks lost 3

Top 100: **15%** got it right · Campus: **4%** (3 of 78)

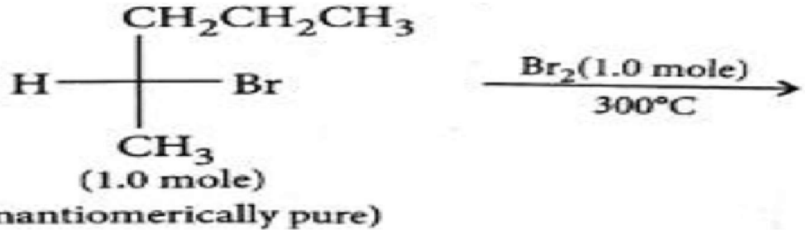
Tough one

You: Wrong

PARAGRAPH-II

Alkanes are aliphatic saturated hydrocarbons and have the general molecular formula C_nH_{2n+2}. The simplest one is methane (CH_4), is also the most abundant. Large amounts are present in atmosphere, in the ground, and in the oceans. Alkanes are most reduced forms of organic compounds. Hence, they can be prepared by reduction of any organic compound. Alkanes are prepared by the catalytic hydrogenation of alkenes or alkynes.

51. In the following monobromination reaction, the number of possible chiral products is



Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q47 **Hydrocarbons – Dichlorination isomers of C_5H_{10} (VD 35)**

JEE Advanced · Chemistry · Integer · marking +4 / 0 · marks lost 4

Top 100: **7%** got it right · Campus: **5%** (4 of 78)

Tough one

You: Wrong

47. A hydrocarbon [X] is having vapour density 35 and it forms only 1 monochloro product under sunlight. How many isomeric products (including stereo) are possible when [X] undergoes dichlorination?

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

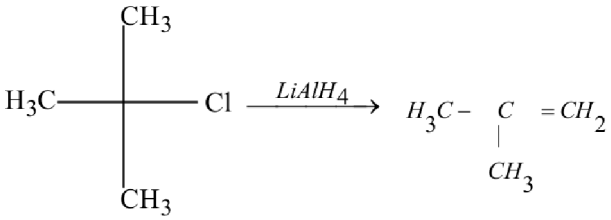
PARAGRAPH-II

Alkanes are aliphatic saturated hydrocarbons and have the general molecular formula C_nH_{2n+2}. The simplest one is methane (CH_4), is also the most abundant. Large amounts are present in atmosphere, in the ground, and in the oceans. Alkanes are most reduced forms of organic compounds. Hence, they can be prepared by reduction of any organic compound. Alkanes are prepared by the catalytic hydrogenation of alkenes or alkynes.

50. Find the No of correct statements out of the following

A) Alkanes are prepared by catalytic hydrogenation of Alkenes where both hydrogen atoms add to the same face of the alkene (syn-addition).

B) Alkyl halides when reacts with Lithium Aluminium Hydride form corresponding alkane: $R-X \xrightarrow{LiAlH_4} R-H$ for R : methyl , $1^0, 2^0$

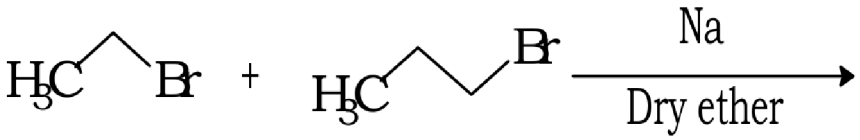


C)In case of Tertiary Alkyl halide is observed.

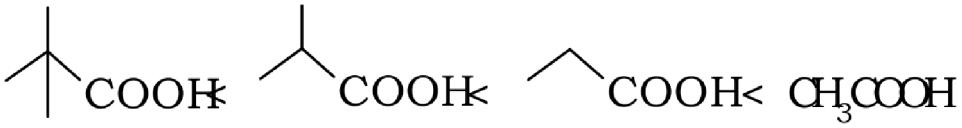
D) Grignard's reagent when treated with a polar protic solvent gives alkane:

$R-Mg+HA \rightarrow R-H+MgX(A)$

E) The different compounds obtained by the following reaction are 7 (only organic compounds)



F) Rate of sodalime decarboxylation for the following acids follow:



G) The side products of potassium butanoate during Kolbe's Electrolysis are propane, propene and propyl butanoate

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

10 How to read this report

Where the numbers come from. Marks, percentages, All-India / State / Campus ranks, subject marks, subject All-India ranks and the wrong / skipped questions are the official results in the institute's *All India Marks Analysis* (graded with the final answer key). Maximum marks leave out deleted questions — JEE Mains: Q58 — JEE Advanced: Q45. **Program rank** = rank in Pragya G9 (EXT) Achievers across all campuses, from the batch's question-wise sheet († JEE Advanced: that sheet was graded before the final key, so this rank may differ slightly). **Top 100** = the batch's top 100 across all campuses, from the same question-wise sheet; * where it has no data for a question, the campus figure decides the priority. **Campus** = all students of this program on campus. **Priority:** Easy miss = 70%+ of the top 100 got it right; Within reach = 40–69%; Tough one = below 40%. **Marks lost** on a question = its marks plus the negative mark if answered wrong, its marks if skipped. In JEE Advanced, 3 marks went on multiple-correct answers that were only partly correct. **Weekly-test average** = your average over the weekly Mains tests (WTM) or Advanced tests (WTA) before this Part Test, from the campus averages sheet. The weekly Advanced tests are marked out of 240; their averages are scaled to 180 to match this paper (× 180/240, so 80 per subject becomes 60). Percentages are unchanged.